

Side canyon

Side canyons are carved by tributaries of the main river channel. Those of the Grand Canyon have contributed to its width.

◁ INCISED MEANDER

Meanders are winding bends usually found in the flat lower courses of rivers. Those occurring on an entrenched river—one that is confined to a canyon—are known as incised meanders. These cut deep gorges, like those seen in the Grand Canyon, along preexisting routes.

**△ KAIBAB PLATEAU**

The Grand Canyon cuts through the southern end of the Kaibab Plateau, which is capped with limestone.

The North Rim

More elevated than the South Rim, the North Rim experiences much colder temperatures and snowfall during winter.

Steep cliffs

Cliffs are often made of more durable rocks, usually strongly jointed sandstone and limestone.

Rock layers

Sedimentary rock strata are usually laid down in horizontal beds, with younger layers above older layers. This creates a geological record that is revealed through erosion.

Oldest rock layer

Unlike the sedimentary rocks above, the oldest rock layers are made up of igneous and metamorphic rocks. These hard rocks resist weathering and erosion, so the bottom of the canyon is narrow with steep sides.

Colorado River

The Colorado flows along the floor of the canyon and continues to cut through layers of rock.

▷ VISHNU BASEMENT ROCKS

These very hard, crystalline rocks—a combination of schist, granite, and gneiss—exposed near the bottom of the Grand Canyon are among its oldest rocks, having formed more than 1.7 billion years ago.

